

SANAPALA SANJAY

Roll No.: U24AI037

Bachelor of Technology

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GitHub Profile

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EDUCATION

•Bachelor of Technology in Artificial Intelligence

2024-28

Sardar Vallabhbhai National Institute of Technology

CGPA: 8.92

PERSONAL PROJECTS

•LifeFlow — Personal Management Web App

A full-stack Django-based productivity and personal management platform integrating multiple daily-use tools into a unified dashboard.

- Developed an all-in-one dashboard integrating To-Do List, Expense Tracker, Notes App, Calendar, and Inventory Management system.
- Implemented secure user authentication using Django's built-in authentication system for signup/login functionality.
- Designed and structured scalable backend models, views, and forms using Django MVC architecture.
- Built dynamic UI using Django Templates with HTML and CSS for responsive user interaction.
- Managed database operations using SQLite including migrations and schema design.
- Technology Used: Django, Python, HTML, CSS, SQLite.

•Prompt Injection Defense Gateway

A multi-layered security gateway protecting LLM applications against prompt injection, jailbreak, and multi-turn social engineering attacks.

- Engineered a 3-layer defense architecture combining Regex-based detection, ML-based classification (DeBERTa v3), and LLM-based semantic judgment (Google Gemini).
- Implemented preprocessing pipeline for Unicode normalization, encoding/decoding (Hex, Base64), and homoglyph detection to prevent obfuscation attacks.
- Built semantic intent detection using sentence-transformer embeddings with cosine similarity scoring to detect paraphrased attacks.
- Designed a dynamic policy engine supporting actions: ALLOW, SANITIZE, REPROMPT, and BLOCK with adaptive risk thresholds.
- Developed real-time attack simulation dashboard with Shield ON/OFF toggle using React and Vite.
- Integrated FastAPI backend with PyTorch, HuggingFace Transformers, Groq (Llama 3.1), and Gemini APIs.
- Technology Used: FastAPI, PyTorch, Transformers, Sentence-Transformers, React 19, Vite, Gemini API, Groq API.

TECHNICAL SKILLS AND INTERESTS

Languages: C/C++, Python, Javascript, HTML+CSS

Libraries : C++ STL, Python Libraries, ReactJs

Web Dev Tools: Nodejs, VScode, Git, Github

Frameworks: ReactJs

Cloud/Databases: MongoDB, SQLite, Relational Database(mysql)

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object Oriented Programming, Database Management System, Artificial Intelligence.

Areas of Interest: Web Design, Artificial intelligence, Machine Learning.

Soft Skills: Problem Solving, Self-learning, Adaptability

POSITIONS OF RESPONSIBILITY

• Active BIS member

2025-28

• Active Nexus member

2024-28